

Kunming AI+

An Applied Introduction to AI with PyTorch

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Session Title	Topics
Workshop 1	VS Code, Conda and PyTorch
Practical Workshop 1	Constructing and training a neural network from scratch
Workshop 2	PyTorch modules for building and training neural networks
Practical Workshop 2	PyTorch modules for data loading
Workshop 3	Diagnosing training performance
Practical Workshop 3	Initialisation and regularisation
Academic Lecture 1	Physics Informed Neural Networks
Academic Lecture 2	Neural Operators
Workshop 4	Convolutional neural networks
Practical Workshop 4	Scikit Learn
Workshop 5	Complete end-to-end experiment
Practical Workshop 5	Open Q&A

Abstracting the data

Last session we abstracted the model, the loss, and the optimiser.

However, we still kept the whole dataset in one tensor and fed to the training loop in at once.

That's fine for a toy problem. It breaks the moment data is bigger than memory, needs shuffling and batching each epoch, or needs preprocessing on the fly.

Today we build the standard PyTorch input pipeline with `Dataset` and `DataLoader`.

The whole idea

PyTorch Dataset

Defines one sample.

Provides a method to find a specific sample, and compute the total samples

PyTorch DataLoader

Defines a batch.

Wraps a Dataset and hands you shuffled, batched, parallel-loaded tensors.

Train / validation / test

Before creating any datasets or data loaders, we first split the data into three datasets.

Train

~60%

The model learns from this data.

Validation

~20%

Model is evaluated on this during training to tune hyperparameters.

Test

~20%

Evaluated on once, at the very end of training, to estimate real-world performance.

Full dataset

When the data doesn't fit in memory

Loading everything up front stops working once the dataset is bigger than RAM.

Streaming

Read samples one at a time, in order.
Never holds the whole set, but you can't index or shuffle it.

Useful for: files, logs, live data

Memory-mapping

The file stays on disk; the OS loads only the row you ask for. Indexing and shuffling still work.

Useful for: large arrays on disk

Bigger still? HDF5 and Parquets. Not covered in this course.

Time for some code...